Performance-Driven MAC Protocol Design for Next-Generation Wireless Systems

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1 Introduction

Next-generation wireless systems require high efficiency, low latency, and intelligent resource management.Next-generation wireless systems, including advanced fifth-generation (5G) and emerging sixth-generation (6G) networks, are transforming the global communication landscape by enabling ultra-reliable, low-latency, and high-capacity connectivity. The increasing proliferation of Internet of Things (IoT) devices, intelligent transportation systems, augmented reality platforms, and mission-critical industrial applications has imposed stringent performance requirements on wireless communication infrastructures. These requirements include ultra-low latency, high reliability, massive device connectivity, and improved spectral efficiency [2].

Within the protocol stack of wireless communication systems, the Medium Access Control (MAC) layer plays a critical role in determining how multiple devices share limited wireless spectrum resources. The MAC layer directly influences network throughput, delay performance, energy efficiency, fairness, and reliability. In conventional wireless systems, MAC protocols were primarily designed for moderate traffic loads and best-effort services. However, next-generation systems must support heterogeneous traffic patterns, dense deployments, and diverse Quality of Service (QoS) constraints [3]. [8] [11] [8, 11, 12, 14, 4, 9, 7, 16, 10] [13, 6, 15, 17, 5, 1]

Performance-driven MAC protocol design focuses on optimizing measurable performance indicators such as latency, throughput, packet delivery ratio, and energy consumption. Instead of relying solely on static scheduling or random access strategies, modern MAC designs incorporate adaptive resource allocation, intelligent scheduling, traffic prioritization, and learning-based optimization mechanisms. These approaches aim to dynamically respond to network conditions and application requirements.The commercialisation of 5G in the sub-6GHz (FR1) spectrum began in 2018 while operation in the mmwave (FR2) spectrum has already been standardized [28] and will be deployed in the years to follow. The first phase of 5G was released as a Non-Standalone deployment, or 5G-NSA, in which CP signalling is realised in the 4G infrastructure while UP takes advantage of the better DL/UL 5G performance. 4G or 5G architecture can be used at the core depending on the 5G-NSA deployment. The standalone implementation of 5G, or 5G-SA, won't rely on the previous generation. A combination of a 5G core and 5G New Radio (NR) will fully enable the capabilities of the next generation of mobile networks. The three main use-case categories for 5G as defined by 3GPP are: enhanced Mobile Broadband (eMBB), massive Machine-Type-Communications (mMTC) and Ultra Reliable Low Latency Communications (URLLC). The scarcity of the available microwave spectrum, which consists of a wireless communications sweet-spot, led the mobile communications community to consider the utilization of higher fre quency bands. The 2015 World Radio Conference decided the investigation of mmWave frequencybands in 5G and beyond networks for frequencies ranging from 24 GHz to 86 GHz. A standard for the mmwave spectrum was first introduced as FR2 with 3GPP Rel.15 (2017), which specified bands n257 to n261. The FR2 bands range between 24.25 GHz and 52.6 GHz and offer a much wider resource availability than FR1 bands. 3GPP continues to investigate other mmwave fre quency options to allocate more resources to 5G networks. For example, the spectrum at 60GHz 80GHz is under consideration along with an unlicensed spectrum access approach referred to as 5G NR-U. Although the mmWave spectrum is very promising, additional work is required to take advantage of its full potential. The propagation characteristics of the mmwave radiation differ to the ones of microwaves. Nevertheless, the mobile communications community has been pushing towards new advancements that can overcome the mmwave challenge and let mobile networks take advantage of the wide resource availability at higher frequencies

This chapter presents a comprehensive exploration of performance-driven MAC protocol design principles for next-generation wireless systems. It discusses conceptual foundations, core methodologies, comparative perspectives, practical use cases, implementation challenges, and emerging research directions. The discussion emphasizes how advanced MAC mechanisms can enable reliable and scalable communication in future wireless ecosystems. As shown in Figure 1, modern MAC design integrates intelligence across multiple layers.

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2 Background and Conceptual Foundations

Wireless communication systems are evolving toward multi-layer optimization and adaptive control mechanisms.The evolution of wireless communication systems has significantly influenced MAC protocol development. Early wireless networks primarily relied on contention-based access schemes such as Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA), which were suitable for low-density environments. As network complexity increased, structured approaches such as Time Division Multiple Access

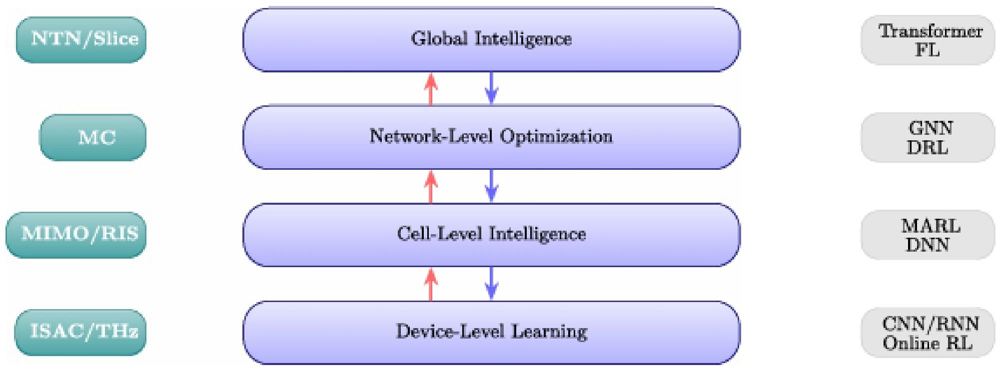

Figure 1: Hierarchical AI-Driven MAC Intelligence Framework

(TDMA) and Orthogonal Frequency Division Multiple Access (OFDMA) were introduced to enhance spectrum utilization and reduce collisions.In order to meet the extremely high capacity and throughput demands, wider frequency bands in the millimeter scale have been considered and standardized for future 5G networks and beyond. In the past few years research [4, 5, 27, 29?–53] has been focusing on new challenges related to mmwave channel modeling, network design and integration, radio technologies and the deployment of MIMO and Massive-MIMO (M-MIMO) antennas. In the sub-6GHz spectrum, Mobile Network Operators (MNOs) have been sharing licensed or unlicensed spectrum as a way to expand their available resources and achieve better network performance. With the introduction of 3GPP Rel.15, 5G is set to take advantage of all resources 6 1.4. THESIS OUTLINE AND CONTRIBUTIONS allocated to mobile communications and implement a multi-spectrum resource utilization by sharing the 4G and 5G sub-6GHz with mmwave bands. Despite the multi-spectrum resource sharing, multi-operator spectrum access must also be challenged by taking into consideration the unique propagation characteristics at mmwave frequencies and the much wider bandwidths. Exclusively licensing the mmwave spectrum cannot be justified given the natural mitigation of interference at such frequencies. Partially or fully sharing the available mmwave spectrum could enable MNOs to significantly improve the performance of their networks by using the available resources more efficiently in terms of spectral occupancy. Various studies [13, 15, 18, 19, 54–63] have been investigating partially or fully sharing the mobile communications spectrum along with the network infrastructure, even at sub-6GHz. Other techniques have also been suggested to have a positive effect in spectral occupancy and overall network performance such as cell sectorization [22] and () [64–70] also has the potential to significantly improve the network performance by aggressively re-using the spectrum. However, highly aggressive resource utilisation methods combined with ultra-dense cellular deployments are expected to introduce additional network challenges in terms of inter-cell and intra-cell interference. This thesis addresses the following research questions: to what extend can mmwave networks provide an operational environment for such methods, what are the actual performance expectations of mmwave mobile networks under SS and what are their limits

2.1 Evolution of Wireless Network Generations

Fourth-generation (4G) systems introduced improved spectral efficiency and broadband capabilities. However, 5G networks expanded the scope to support enhanced mobile broadband (eMBB), ultra-reliable low-latency communication (URLLC), and massive machine-type communication (mMTC). These service categories require differentiated MAC strategies to accommodate varying traffic characteristics and performance constraints.

Emerging 6G systems are expected to integrate artificial intelligence, terahertz communication, integrated sensing, and ultra-dense deployments. Such environments necessitate MAC protocols capable of predictive scheduling, cross-layer optimization, and adaptive control mechanisms.the reliability of our results and conclusions as well as providing data for comparison with the theoretical results. In Chapter 4, the performance of SS in a large-scale, multi-operator mmwave cellular network [71] was evaluated using combination of ray tracing with a real-world 3D map database and a realistic multi-element antenna deployment with directional beamforming. The performance metrics in the SS environment were compared to the Exclusive License (EL) model and conclusions were derived about the benefits, feasibility and challenges of SS. To what extend SS combined with a cell sectorization strategy is beneficial or detrimental was investigated in Chapter 5 [72]. Similar to the previous study, the highly aggressive spectrum utilization schemes were ray-traced in a real-world environment simulator. Sectorization and spectrum sharing have proven to be beneficial for the overall throughput performance of a network as standalone spectrum utilization improvement strategies. However, their effectiveness within a spectrum-aggressive environment has yet to be investigated. Following a more realistic approach in terms of channel, antenna and environment modeling, sectorization was deployed in a dense multi-operator mmwave network, the performance of which was evaluated under the EL and SS models. In Chapter 6 [73], a novel beamforming-based technique was introduced for non-orthogonal resource allocation in mmwave networks. An aggressive spectrum utilization strategy incorporat ing SS and NOMA was investigated and a study was conducted on the co-existence of SS and NOMAinamulti-operator network. Again, by taking advantage of a highly realistic engine in terms of channel, antenna and environment modeling, EL and SS instances of the network were simulated and their corresponding performances were evaluated under different antenna and NOMAconditions. The SS network performance was analyzed in terms of Signal-to-Interference and-Noise-Ratio (SINR) and throughput, and was then compared to other schemes such as the FDMAandsectorization. Finally, an interference mitigation strategy was implemented based on a realistic data collec tion provided from the network UEs. The random allocation of resources that was initially implemented in Chapters 5 and 6 was optimized using measurements taken by neighbouring, non-serving UEs. By doing so, the optimization algorithm tried to minimise the interference caused by each sector to the interacting neighbouring UEs by avoiding the allocation of high interfering resources. The same instances of a highly aggressive spectrum re-utilization network were compared with and without the interference mitigation mechanism, and the corresponding network performances were evaluated according to the SINR and throughput gainsray-tracing a real world environment (Chapter 2), while utilizing an antenna model based on measurements of a real patch antenna (Chapter 3). A dense cellular deployment with hundreds of cells was simulated with thousands of UEs connected to a multi-operator network. However, models of upper network layers were not implemented. Exclusively licensing the available spectrum to operators has been known to provide highly reliable access to the network while providing an acceptable trade-off between spectral efficiency and interference. Despite the new interference challenges involved in spectrum sharing, a mmwave implementation could be very beneficial [19] as the beamforming capability at mmwave frequencies can effectively work as an interference mitigation mechanism. Thus, much higher link capacities and data rates are possible. Early research results suggest that SS throughput performance gains can reach 130is not to provide raw network performance metrics, but an accurate comparison between the spectrum access models under investigation and quantify the advantages or disadvantages of SS relative to a conventional spectrum access strategy

2.2 Role of the MAC Layer in Wireless Architecture

The MAC layer is responsible for coordinating access to the shared wireless medium. It performs scheduling, resource allocation, collision management, retransmission control, and traffic prioritization. The effectiveness of MAC protocols directly impacts system-level performance metrics including throughput, latency, fairness, and energy efficiency.

In centralized architectures such as cellular networks, the MAC layer often operates under base station coordination. In distributed networks, devices independently negotiate channel access. Performance-driven design must consider both centralized and distributed coordination models.mmWavefrequencies pose certain physical challenges for wireless communications. The prop agation of mmWaves has been extensively studied by many organizations such as the Federal Communications Commission (FCC), USA as early as in 1997 [33]. Even though signals at lower frequencies can achieve longer propagation range, mmwaves are characterized by very short travel distances. There are various factors affecting the propagation of mmwaves [36] including the atmospheric gaseous attenuation (water vapour absorption, oxygen absorption, precipitation attenuation), rain, foliage blockage, scattering effects, diffused reflections, scattered reflections, and diffraction or bending. Large scale-fading represents the signal-power attenuation or path loss due to motion over large areas [74]. According to [34], the main physical challenge for wireless mmwavenetworks is the strong attenuation over long distance transmissions, which includes two components: Free Space Path Loss and Propagation Losseslosses and blockage probability which result in a limited operational range in practical cellular environments. Despite not being able to provide long range coverage and penetrate objects, the mmwave propagation characteristics could be advantageously exploited by deploying denser cellular deployments and more aggressive spectrum utilization strategies. The short range of mmwave radiation could solve one of the main challenges in wireless communi cations: interference. The natural interference mitigation characteristics of mmwave frequencies such as the high atmospheric losses, rain attenuation, scattering, diffraction and foliage losses offer an ideal environment for new, performance-centric cellular implementations. The available mmwave spectrum can be used more aggressively and non-orthogonally while also allowing denser cellular deployments, significantly improving the spectrum occupancy in wireless mobile networks. The fundamental differences between mmwave and microwave networks introduce challenges with respect to conventional design constraints, objectives, and available degrees of freedom. Exclusive spectrum licensing, orthogonal resource allocation and conventional frequency re-use strategies, currently known as the foundation of cellular networks, could become a bottleneck for future mmwave mobile networks. Re-inventing cellular networks as we know them is the main challenge behind a commercially viable mmwave mobile access. The main problems related to mmwave mobile communications were identified in [50] as synchronization, random access, handover, channelization, interference management, scheduling and association. Due to the multi element and multi-beam antenna technologies at mmwaves, it is suggested that more intelligent MAClayer strategies must be developed for mmwave networks to become practically operable. Some key research areas behind successful mmwave cellular deployments are considered among others: MIMO/M-MIMO antenna technologies, aggressive spectrum access models, ultra-dense networks and NOMA schemes The Spectrum Sharing and Exclusive License models were investigated in a multi-operator mobile network covering an area of 0.5 km2 in central Bristol, UK (see Figure 2.4). Our ray tracing tool performs an extensive ray-path search within a predefined vector map environment, as described in detail in Chapter 2. Multiple propagation mechanisms are supported such as terrain scattering, building reflection and scattering, building rooftop and corner diffraction, etc. 3D propagation of mmwave radiation was simulated based on models obtained from frequency measurements as well as LIDAR (Light Detection and Ranging) and terrain scans taken at the investigated location. Given the high reflection losses, ray tracing was limited to 1st and 2nd order reflection rays affected by the underlying terrain, water, and building structures. Atotal of 500 BSs and 9,023 UEs were ray-traced at different locations creating a ray-tracing pool covering most of the simulation environment. BS antennas were assumed to be installed on street lampposts 10m above ground level, a strategy that was initially suggested and investigated in [110]. UE antennas were placed in streets (90taking into consideration the British lifestyle. The BS and UE selection was made from within the pool such that the density requirements were met, while ensuring no-outages throughout the network, i.e. all UEs were located inside their operator's coverage. Cell locations were randomly

2.3 Performance Metrics in MAC Design

Performance-driven MAC protocols are evaluated based on quantitative metrics. Throughput measures the successful data transmission rate over a given time interval. Latency represents the time delay experienced by packets from transmission to reception. Reliability refers to successful packet delivery probability, particularly critical for mission-critical applications. Energy efficiency evaluates power consumption relative to transmitted data volume.

Balancing these metrics involves inherent trade-offs. For instance, increasing reliability through retransmissions may increase latency and energy consumption. Performance-driven design seeks to optimize these metrics collectively rather than independently.The non-geometric stochastic channel models predict the physical propagation parameters of a channel in a stochastic way, i.e. they are solely based on statistical distributions without having any deterministic information about the geometry of the environment. The Geometry based Stochastic Channel Models, or, GSCMs are stochastic channel models that have a limited deterministic information about the geometry of the propagation environment. Single-bounce and multi-bounce scattering models are two well known GSCMs which randomly place the electromagnetic wave scatterers between the transmitting and receiving ends, taking into account simple geometrical considerations [76–81]. The majority of standardised stochastic channel models derived from extensive measurements are geometry-based. Multiple GSCMs are currently being recommended by 3GPP for system evaluation, namely Spatial Channel Model (SCM) [82] and Spatial Channel Model Extended (SCME)[83]. Conventional stochastic and analytical models combine random variables fitted on traces and measurements. Despite their wide adoption in channel analysis [84, 85] and system-level simulations [83, 85], the generality of such models fit poorly in scenarios with very specific environmentsThe unique characteristics of mmwave channels such as the temporally- and spatially-consistent updates of the LoS conditions, and the evolution of the channel Multi Path Components (MPCs), cannot be accurately represented by most stochastic models [86]. The mmwave modelling chal lenges can be accurately addressed by the use of deterministic channel modelling tools such as ray tracers. Ray-tracing tools have been developed to precisely characterize the propagation of radiated signals in specific environments and conditions [87, 88]. A ray-traced channel is modeled in terms of MPCs based on extensive ray-path search from the transmitting end towards the receiving end. mmWave propagation characteristics such as reflection, diffraction, scattering,etc are applied within the specific, ray-traced environment resulting in highly accurate signal readings including angle of arrival, delay, power and interference [89]. Ray tracing tools are also f lexible in terms of experimental modelling as they can be easily paired with other solutions by integrating 3rd-party environment and antenna models. Therefore, an accurate ray-tracing channel model is as accurate as its least accurate component. High accuracy is not only required in the description of the scenario and the mobility model of the communication endpoints, but also in the simulation environment. assigned to operators meeting the minimum inter-site distance restriction of 50m, ensuring low outage probability while maintaining the inter-cell interference at acceptable levels [111]. This study investigates a 4-operator mobile network, emulating the UK's four major mobile network operators. The total cell and UE densities were 300 cells/km2 and 5,414 UE/km2, respectively. The selection of the cell density was based on Nokia's insights [112] as presented in 2017. Taking into consideration the simulated location and the large number of students (27,513 in 2019) and stuff (7,912 in 2019) [113] studying/working at the University of Bristol, a very dense UE deployment was chosen to emulate a daily scenario where the mobile networks are overcrowded. Indoor propagation was not investigated in this study due to the limitations of the ray tracing tool. Additionally, the analysis only considered single captures of the network and did not apply any mobility models for the UEs. UEs located near the map boundaries were not considered in the statistical analysis of the network performance as these represent edge cases due to the lack of interfering sources. Data was only collected from UEs located in the inner area of the mapThe complex multipath ray-traced propagation data are spatially convolved with the radiation patterns of transmit and receive multi-element antenna arrays. As explained in detail in Chapter 3, a single-patch antenna model was scaled to a multi-element antenna model by combining multiple single-patch antennas that are spaced half a wavelength away from each other. For the purpose of the study in this chapter, the antenna array was limited to a maximum physical size. Consequently, the number of elements varied with frequency, i.e. more antenna elements could fit within the predefined array size at 70 GHz than at 26 GHz. As a result, the HPBW of the antenna got proportionally narrower, as shown in Table 4.2. At 26 GHz, 16×16 and 4×4 arrays were configured for the BS and UE antennas, respectively. The corresponding array configurations at 70 GHzwere 32×32 and 8×8.

The generated 3D radiation patterns of the BS and UE antenna configurations at 26 GHz and 70 GHzare shown in Figures 4.15 and 4.16, respectively. The narrower the HPBW, the higher the power concentration is towards specific directions and the smaller the side lobes. Consequently, signal transmission can be optimised according to the location of the receiver, i.e. UE, while generating lower interference due to the lower signal leakage towards unwanted directions. At the same time, the beamforming accuracy of the antenna system becomes more strict. For example, a beamforming error at 26 GHz can be forgiving for up to 6o, while at 70 GHz the erro

2.4 Conceptual Framework for Adaptive MAC Protocols

Modern MAC protocols incorporate adaptability to dynamic network conditions. Adaptive scheduling dynamically assigns time-frequency resources based on traffic demand. Intelligent access control mechanisms predict channel occupancy patterns. Machine learning techniques can estimate interference levels and optimize resource allocation decisions.University of Bristol (UoB) has been developing a channel modelling tool based on ray-tracing [90–95]. The features of this tool have been expanding throughout the years and novel channel modelling techniques have been integrated and validated in numerous research publications and experimental tests. At the time of research, the aforementioned tool was the only all-in-one ray tracer able to provide support for mmwave propagation. The 'virtual' availability of easily accessible locations in central Bristol and around the UoB was also appealing, given the potential of practically testing and validating any simulated scenarios in case mmwave mobile equipment became available at any stage throughout the duration of this research study. Nevertheless, other ray-tracing tools were considered and/or evaluated, but it was deduced that they were either at very early stages with very limited features (e.g. NYUSIM v12) or too costly and not readily available (Remcom's Wireless InSite3). Propagation mechanisms such as terrain scattering, buildings reflection and scattering, buildings rooftop and corner diffraction and foliage loss were supported by the UoB Ray-Tracer among others. The ray-tracing engine implements an exhaustive ray-path search to identify all possible radiation paths between the transmitter and the receiver within the 3D simulation environment. A vector map database provides detailed and high accuracy information about the terrain, building and foliage. The available simulation environments consist of high-accuracy, vector map databases with up to 1m resolution. Building structures were accurately represented, as shown in Figure 2.4. Due to data confidentiality, more details about the latest developments of the software environment of this tool cannot be shared. For the requirements of this study, an extensive ray-path search was implemented in a vector map database of central Bristol, UK, covering an are of 0.5km2. The channel propagation was based on a deterministic model developed from measurements taken at a wide range of frequencies and surfaces within the simulation environment. As a result, environment characteristics such as the material properties of the underlying terrain and building structures were accurately affecting the mmwave propagation of the investigated network scenarios. Given the high penetration and reflection losses at mmwave frequencies, only rays of 1st and

2nd reflection orders were consideredthe direction of departure of the strongest ray while the user antenna is directed towards the corresponding receiving direction. Every BS antenna array is capable of communicating using all channels at any required direction, depending on the channel allocation of the connected users. There are no limitations on the number of concurrent links or the total transmission power requirement (each link is still bound by the EIRP constrain). Immediately surrounding BSs are the only sources of interference considered. In the FDMA environment, no interference is produced by a serving BS to its connected users. For the exclusive license model, all non-serving BSs that belong to the same operator as the user are capable of interfering with each other. In spectrum sharing the same consideration holds true but for all non-serving BS, i.e. including those of other operators. BSs with no connected UE do not transmit any data and are assumed to cause zero interference. Beamforming is applied to all assigned channels independently for every user. The corresponding signal and noise powers are computed on a per-channel basis. It is thus possible for a user to perform better or worse in different channels. Nonetheless, the channel usage remains balanced between all assigned channels. If we consider BSi to be the serving BS of user uk, let Nri (z) be the total number of received signal rays in channel z and vk,i,n(z) be the received power in watts of a signal ray n, n 1,...,Nri after the corresponding BS and user antenna gains have been applied, such that: Nri (The performance of a multi-operator network under the spectrum sharing and exclusive license models was investigated in the 26 GHz and 70 GHz spectra. A channel model based on a more detailed simulation perspective including site specific ray tracing around the university campus was utilized along with antenna modelling generated from measurements of a real patch antenna. While previous works on this topic considered more theoretic assumptions and methodologies, this chapter looked at this problem from a more realistic perspective. Spectrum sharing is beneficial for the majority of users, even without any interference mitigation techniques. While exclusively licensing the spectrum provides better spectral efficiency, it also under-performs in terms of spectral utilization. Spectrum sharing makes a wider spectral use achieving higher link capacities for the majority of the users. As a result, the spectrum sharing network achieved a total capacity of 2 to 3 times higher than the corresponding exclusive license network. By statistically grouping users into 5th, 50th and 90th percentiles based on their achieved capacity, i.e. low, average and high SINR respectively, spectrum sharing performed significantly better in the higher percentiles under ideal beamforming conditions. However, the high interference levels experienced by 5th percentile users resulted in marginal to no benefit at all over the licensed model. Beamforming errors can have a significant impact on the overall network performance, especially when the spectrum is fully shared. Even though 50th and 90th percentile performance gainremained high, beamforming errors significantly affected the low SINR users. As a result, the exclusive license model outperformed spectrum sharing by more than 50Based on our error model and under the assumption of higher beamforming accuracy, narrower antenna beamwidths and the higher propagation losses at 70 GHz made the network less sensitive to beamforming errors. Practical mobile communications networks could experience additional antenna losses and performance degradation, especially when moving targets are involved. Thus, it is possible for interference in a mmwave spectrum sharing network to become a limiting factor due to limitations and imperfections in current antenna technologies and physical layer protocols. Network operators should consider the deployment of high-accuracy MIMO antennas as one of their main interference mitigation strategies. The lower interference levels at 70 GHz did not translate into higher SINR. According to Table 4.5, 26 GHz and 70 GHz networks both had a similar average network performance despite the higher number of antenna elements at 70 GHz. It is evident that the higher propagation losses at 70 GHz result in a relatively lower signal power which in turn results in a reduction in SINR. Adenser cellular deployment and/or a higher EIRP constraint should be considered in order to improve the signal reception towards the higher-end of the mmwave spectrum. Fully sharing the spectrum is unlikely to offer any performance benefits for users with low SINR. The 'outage' threshold in mmwave mobile networks could play an important role in the network performance. Given that the operation in the mmwave spectrum is mostly performance-driven rather than coverage-driven and assuming support of dual-connectivity with the sub-6GHz bands, dropping the low SINR users out of the simulation, i.e. considering them in outage, could be a rather realistic assumption. It is worth noting that SS could have benefited under rainy conditions. As previously explained in Section 2.1.2.2, rain droplets and water vapour can introduce significantly higher propagation losses at mmwave frequencies. Even though this would have resulted in higher outage probability, it would have also caused a reduction in inter-cell interference. Assuming the received interfering rays travel longer distances than the signal rays, the rain attenuation losses of the latter are higher than the losses of the former. As a result, SS gains would have increased relative to the EL model. Further network simulations could benefit from the introduction of water vapour and rain droplets in the channel modelling. Interference mitigation mechanisms accompanied by an overhead estimation function could also provide a better view of the potential of SS. Utilization of multi-element antennas with wider HPBW at the transmitter should also be studied as they could eliminate the need for high-accuracy directional beamforming and provide links of higher reliability.

Figure 2 illustrates layered communication strategies for performance optimization.

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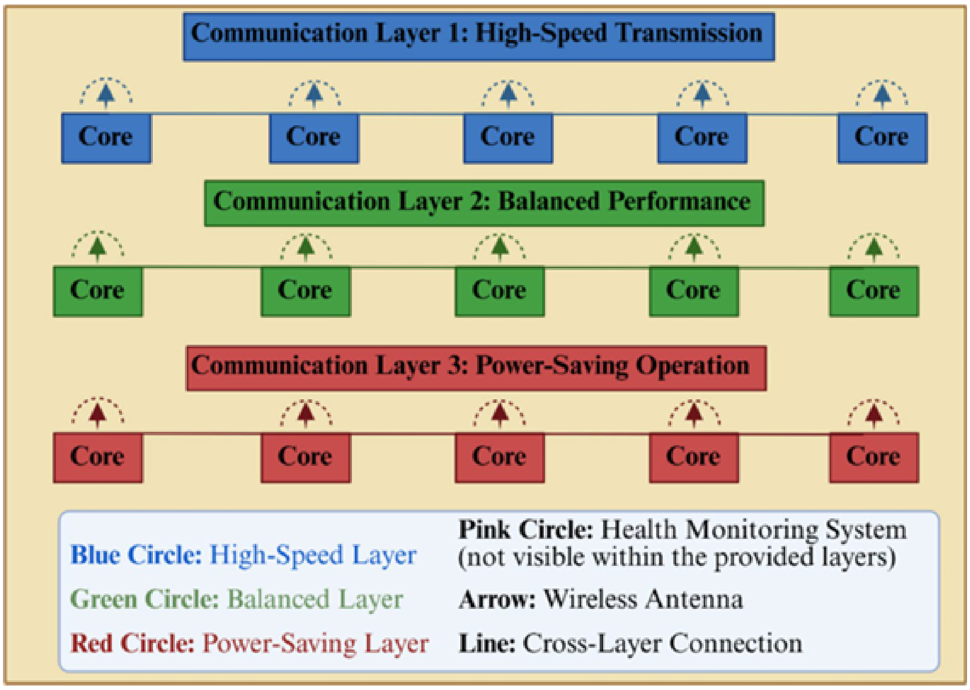

Figure 2: Multi-Layer Communication Architecture for Performance Optimization

3 Core Concepts and Approaches

Interference management and efficient MAC design are critical in dense wireless systems.Performance-driven MAC protocol design integrates several key conceptual approaches. Adaptive scheduling mechanisms dynamically allocate time-frequency resources based on traffic demand and priority levels. These mechanisms reduce latency for delay-sensitive services while maintaining fairness among users.

Dynamic resource allocation strategies utilize real-time channel state information to optimize bandwidth distribution. By considering signal quality, interference patterns, and traffic requirements, these strategies enhance spectral efficiency and reduce packet loss.

Learning-assisted MAC techniques incorporate machine learning algorithms to predict channel occupancy, estimate traffic patterns, and optimize contention windows. Reinforcement learning-based scheduling has demonstrated significant improvements in throughput and reliability in dense wireless environments.

Cross-layer optimization further strengthens MAC performance by enabling interaction between the MAC and physical layers. Information such as channel quality indicators and modulation schemes can inform adaptive access decisions, improving overall system efficiency.

4 Practical Insights and Use Cases

Performance-driven MAC protocols enable diverse real-world applications. In industrial automation systems, ultra-reliable low-latency communication ensures safe coordination between robotic systems and control units. Adaptive scheduling minimizes transmission delay and enhances reliability in such mission-critical scenarios.

In smart city deployments, large numbers of Internet of Things devices generate heterogeneous traffic patterns. Dynamic resource allocation improves spectrum utilization while maintaining energy efficiency for battery-powered devices.

Autonomous vehicle communication systems rely on rapid data exchange between vehicles and infrastructure. MAC protocols optimized for low latency and high reliability reduce collision risks and improve traffic safety.

Augmented and virtual reality applications require high throughput and stable connectivity. Intelligent MAC strategies allocate sufficient bandwidth to support immersive user experiences while maintaining fairness across multiple users.Performance-driven MAC protocol design increasingly relies on adaptive and context-aware decision mechanisms. Unlike traditional static MAC schemes, modern approaches dynamically adjust scheduling policies according to traffic demand, interference patterns, and service requirements. In dense wireless environments, static allocation often results in inefficient spectrum utilization and unfair access distribution. Adaptive mechanisms overcome this limitation by continuously monitoring network state information and reconfiguring access parameters accordingly.

One major approach involves dynamic resource allocation based on channel state information. By evaluating signal-to-interference-plus-noise ratios and traffic queue lengths, the MAC layer can assign time-frequency resources more efficiently. This enhances spectral efficiency while reducing packet delay. In ultra-reliable low-latency communication scenarios, resource pre-allocation techniques may be employed to guarantee deterministic transmission opportunities for critical traffic flows.

Learning-based MAC protocols represent another emerging paradigm. Reinforcement learning algorithms enable the MAC layer to learn optimal channel access strategies through interaction with the environment. By observing collision rates and throughput outcomes, the system gradually converges toward efficient contention window configurations. Such approaches are particularly beneficial in highly dynamic environments where traffic patterns are unpredictable.

Cross-layer optimization further strengthens performance-driven MAC design. Information exchange between the physical layer and MAC layer enables more informed scheduling decisions. For instance, adaptive modulation and coding schemes at the physical layer can influence MAC scheduling priorities. By jointly optimizing these layers, overall system throughput and reliability can be significantly improved.

Mathematical modeling of MAC optimization often involves multi-objective formulations. Throughput maximization, latency minimization, and energy efficiency improvement are typically treated as competing objectives. Weighted optimization techniques allow prioritization of mission-critical traffic while maintaining fairness across users. Such formulations provide a systematic framework for evaluating protocol performance.

Extended Comparative Discussion

Traditional contention-based MAC protocols such as CSMA/CA offer simplicity and decentralized control. These protocols perform adequately in low-density networks but experience significant degradation under heavy traffic conditions. Collision probability increases exponentially with node density, resulting in retransmission overhead and increased latency.

Scheduled access protocols, including TDMA-based systems, provide deterministic delay guarantees and improved fairness. These mechanisms are well suited for centralized cellular networks where a base station coordinates resource allocation. However, purely scheduled approaches may suffer from underutilization when traffic demand fluctuates unpredictably.

Hybrid MAC schemes combine contention-based and scheduled mechanisms to balance flexibility and reliability. For example, control traffic may use scheduled slots while best-effort data employs opportunistic access. This dual-mode strategy enhances resource utilization while preserving quality-of-service guarantees for critical applications.

AI-driven MAC protocols introduce adaptability beyond traditional algorithmic approaches. By leveraging predictive analytics, these systems anticipate traffic surges and proactively allocate resources. Although such techniques improve efficiency, they introduce computational complexity and require robust training data management. Comparatively, adaptive and hybrid models demonstrate superior performance in next-generation wireless environments characterized by heterogeneous traffic and ultra-dense deployments.

Extended Practical Insights and Use Cases

In industrial automation environments, robotic coordination requires ultra-low latency and near-perfect reliability. Performance-driven MAC protocols allocate prioritized transmission slots to control signals while minimizing retransmission delays. This ensures safe and synchronized operation of industrial machinery.

Smart city infrastructures involve large-scale deployment of sensors and IoT devices with diverse communication requirements. Energy-efficient MAC mechanisms extend battery life of low-power devices while maintaining network stability. Adaptive duty-cycling and traffic-aware scheduling are essential for supporting large-scale connectivity without excessive energy consumption.

Autonomous vehicle communication demands rapid exchange of safety-critical information between vehicles and infrastructure. Performance-driven MAC strategies reduce collision probability and ensure deterministic delay bounds. Intelligent scheduling combined with edge computing integration enhances responsiveness in high-mobility scenarios.

Immersive applications such as augmented and virtual reality require sustained high throughput and minimal jitter. MAC protocols optimized for bandwidth aggregation and interference management enable smooth multimedia streaming. Dynamic resource allocation ensures fairness among users while preventing network congestion.

Extended Challenges and Open Issues

Despite significant advancements, several unresolved challenges remain. Ultra-dense network environments introduce severe interference management complexity. Coordinating access among thousands of devices requires scalable algorithms capable of operating with limited signaling overhead.

Balancing latency, throughput, reliability, and energy efficiency remains an inherently multi-objective problem. Optimizing one metric often degrades another. Designing algorithms that dynamically adapt to shifting service priorities without compromising system stability is a major research challenge.

Security concerns also arise in intelligent MAC frameworks. Adversarial interference, spoofing attacks, and malicious traffic manipulation can disrupt learning-based scheduling mechanisms. Ensuring robustness against such threats requires integration of secure authentication and anomaly detection systems.

Standardization poses additional complexity. Next-generation wireless systems involve heterogeneous technologies, including reconfigurable intelligent surfaces, millimeter-wave communication, and edge intelligence. Achieving interoperability across diverse hardware platforms demands unified protocol frameworks.

Extended Future Directions

Future MAC protocol design is expected to integrate artificial intelligence at a foundational level. AI-native networks will employ predictive models that anticipate congestion and dynamically reshape scheduling policies in real time. Federated learning may enable distributed intelligence without compromising user privacy.

Integration of terahertz communication will introduce new challenges in beam alignment and directional access control. MAC protocols must evolve to manage highly directional links and rapidly changing propagation conditions.

Reconfigurable intelligent surfaces will further transform channel characteristics by enabling programmable radio environments. MAC coordination mechanisms must account for dynamic surface configurations and adaptive reflection patterns.

Edge computing will support decentralized MAC intelligence, reducing latency and improving scalability. Localized decision-making at edge nodes can significantly enhance responsiveness in mission-critical applications.

Quantum-resistant security protocols may also influence future MAC architectures as communication systems become increasingly integrated with sensitive infrastructure.

5 Challenges and Open Issues

Despite significant advancements, several challenges remain in performance-driven MAC protocol design. Ultra-dense network deployments introduce severe interference management issues, complicating channel access coordination.

Balancing latency, throughput, and energy efficiency remains a complex optimization problem due to inherent trade-offs among these metrics. Designing scalable algorithms capable of supporting billions of connected devices poses additional technical difficulties.

Security and privacy concerns also emerge in intelligent MAC frameworks that rely on data-driven decision-making. Malicious interference and adversarial attacks may degrade performance and compromise system integrity.

Standardization challenges further complicate implementation, as interoperability across diverse devices and vendors must be ensured. These open issues motivate continued research into robust, adaptive, and secure MAC mechanisms.Performance-driven MAC protocol design requires rigorous analytical modeling to predict system behavior under varying traffic and network conditions. Analytical models provide insight into collision probability, delay distribution, and throughput performance without requiring extensive simulation. Queueing theory has traditionally been used to analyze contention-based MAC protocols, where each node is modeled as a stochastic process with packet arrival and service rates.

In dense networks, Markov chain models are frequently applied to represent backoff mechanisms and retransmission processes. These models enable derivation of steady-state transmission probabilities and collision rates. By analyzing equilibrium conditions, designers can determine optimal contention window sizes that maximize throughput while minimizing delay. Such mathematical modeling supports parameter tuning for adaptive MAC algorithms.

Stochastic geometry has also emerged as a powerful tool for modeling large-scale wireless networks. By representing nodes as spatial point processes, it becomes possible to derive probabilistic interference and coverage models. These analytical tools are particularly relevant for ultra-dense 5G and 6G deployments, where spatial interference patterns significantly affect MAC-layer decisions.

Performance evaluation increasingly incorporates risk-sensitive metrics. Instead of considering only average latency, modern frameworks analyze tail latency distributions to ensure reliability in mission-critical applications. Ultra-reliable low-latency communication scenarios demand extremely low outage probabilities, which requires MAC mechanisms capable of deterministic or semi-deterministic access control.

Energy-Aware MAC Protocol Optimization

Energy efficiency is a primary concern in next-generation wireless systems, particularly for massive IoT deployments. Battery-powered devices require MAC protocols that minimize idle listening, retransmissions, and unnecessary signaling overhead. Traditional duty-cycling approaches reduce energy consumption by allowing nodes to periodically enter sleep states. However, static duty cycles may cause synchronization issues and increased latency.

Performance-driven energy-aware MAC protocols dynamically adjust sleep schedules based on traffic intensity and priority levels. Predictive models estimate packet arrival patterns and activate nodes only when transmission opportunities are likely. This significantly reduces idle power consumption while maintaining acceptable delay bounds.

Energy harvesting technologies introduce additional complexity. Devices powered by renewable energy sources such as solar or kinetic energy must adapt transmission schedules based on harvested energy availability. MAC protocols designed for energy-harvesting networks incorporate energy-aware scheduling to prevent battery depletion while ensuring reliable communication.

In 6G environments, intelligent power control at the MAC layer may coordinate with physical-layer beamforming strategies to reduce overall energy expenditure. By combining power adaptation with traffic-aware scheduling, networks can achieve sustainable long-term operation without sacrificing throughput.

MAC Protocol Design for Ultra-Reliable Low-Latency Communication

Ultra-reliable low-latency communication represents one of the most demanding service categories in next-generation wireless systems. Applications such as remote surgery, industrial robotics, and autonomous driving require end-to-end latency on the order of milliseconds with extremely high reliability.

To meet these requirements, MAC protocols incorporate grant-free access schemes and mini-slot scheduling. Grant-free transmission reduces signaling overhead and eliminates scheduling request delays. However, it increases collision probability if not carefully managed. Performance-driven approaches mitigate this issue through preconfigured resource allocation and redundancy strategies.

Hybrid automatic repeat request mechanisms enhance reliability by combining error correction coding with retransmission protocols. Time-sensitive traffic may utilize parallel transmission paths to reduce outage probability. Deterministic scheduling, where critical packets are assigned fixed transmission slots, further guarantees bounded delay.

These techniques collectively enable predictable communication performance in latency-sensitive scenarios. Nevertheless, ensuring scalability under massive device connectivity remains a significant research challenge.

Spectrum Sharing and Coexistence Mechanisms

Next-generation wireless systems operate across diverse frequency bands, including sub-6 GHz, millimeter-wave, and potentially terahertz frequencies. Efficient spectrum sharing is essential to maximize utilization and prevent interference. MAC protocols must coordinate access among heterogeneous devices operating under different standards.

Licensed and unlicensed spectrum coexistence introduces additional complexity. For example, 5G systems operating in unlicensed bands must coexist with Wi-Fi networks. Performance-driven MAC protocols incorporate listen-before-talk mechanisms and adaptive channel selection strategies to minimize cross-technology interference.

Dynamic spectrum access frameworks allow devices to opportunistically utilize underutilized frequency bands. Cognitive MAC protocols detect spectrum holes and allocate resources accordingly. Such mechanisms enhance spectral efficiency while maintaining fairness across competing systems.

Future wireless systems may rely on distributed spectrum marketplaces, where spectrum allocation decisions are influenced by demand-based pricing and dynamic negotiation. MAC protocols must integrate economic and regulatory considerations into access coordination mechanisms.

Integration with Edge Intelligence and Cloud-Native Architectures

The transition toward cloud-native and edge-enabled wireless architectures significantly impacts MAC protocol design. Edge computing platforms provide localized processing capabilities that reduce latency and enable real-time analytics. MAC-layer intelligence can be partially offloaded to edge nodes, enabling faster scheduling decisions.

Cloud-native architectures allow virtualization of MAC functions through software-defined networking principles. Virtualized baseband processing enables flexible resource allocation and dynamic reconfiguration. Performance-driven MAC protocols can leverage network slicing concepts to allocate dedicated resources for different service categories.

Distributed learning techniques, including federated learning, allow multiple edge nodes to collaboratively train scheduling models without centralized data aggregation. This approach enhances privacy while maintaining system-wide performance optimization.

Integration with programmable network infrastructures also enables real-time policy updates. MAC scheduling rules can be dynamically adjusted based on traffic analytics and service-level agreements. Such flexibility is essential for supporting heterogeneous applications in 6G ecosystems.

Reliability Modeling and Statistical Quality-of-Service Guarantees

Beyond average performance metrics, statistical quality-of-service guarantees are increasingly emphasized. Instead of simply optimizing mean delay, MAC protocols must ensure that the probability of exceeding a latency threshold remains below a specified target. This probabilistic approach aligns with reliability requirements of industrial automation and autonomous systems.

Effective capacity theory provides a framework for analyzing delay-constrained communication systems. By modeling the maximum constant arrival rate that satisfies statistical delay constraints, designers can evaluate protocol suitability for mission-critical applications.

Redundancy-based transmission strategies also improve reliability. Multi-connectivity allows devices to transmit duplicate packets through multiple access points. While this increases resource consumption, it significantly enhances robustness against link failures.

Future research may explore adaptive redundancy schemes that activate additional transmission paths only when channel conditions deteriorate. Such strategies balance reliability and resource efficiency.

Scalability Considerations in Massive IoT Environments

Massive machine-type communication scenarios involve billions of interconnected devices. MAC protocols must scale efficiently without excessive signaling overhead. Random access channel congestion is a significant issue in massive IoT deployments, particularly during synchronized reporting events.

Group-based scheduling mechanisms reduce signaling overhead by aggregating devices into clusters. Cluster heads coordinate access within groups, reducing contention at the base station. Hierarchical MAC architectures enable scalable coordination across multiple network tiers.

Load balancing techniques distribute traffic across multiple frequency bands and access points. Adaptive clustering further enhances scalability by dynamically reorganizing device groups based on traffic patterns.

Ensuring fairness among heterogeneous devices remains challenging. Performance-driven MAC protocols must prevent starvation of low-priority traffic while guaranteeing resources for critical services

Figure 3 highlights interference challenges and MAC-based mitigation techniques.

Performance-driven MAC protocols incorporate adaptive scheduling, intelligent resource allocation, and learning-based optimization mechanisms to enhance network performance.

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6 Comparative Discussion

Different MAC protocols exhibit varying performance characteristics depending on network conditions.Traditional contention-based MAC protocols offer simplicity and low implementation complexity. However, they often experience performance degradation under high traffic density due to increased collision probability and retransmission overhead.

Scheduled access protocols provide deterministic delay guarantees and improved fairness. These mechanisms are particularly effective in centralized cellular systems where base stations coordinate resource allocation. Nevertheless, they may introduce higher signaling overhead and reduced flexibility in highly dynamic environments.

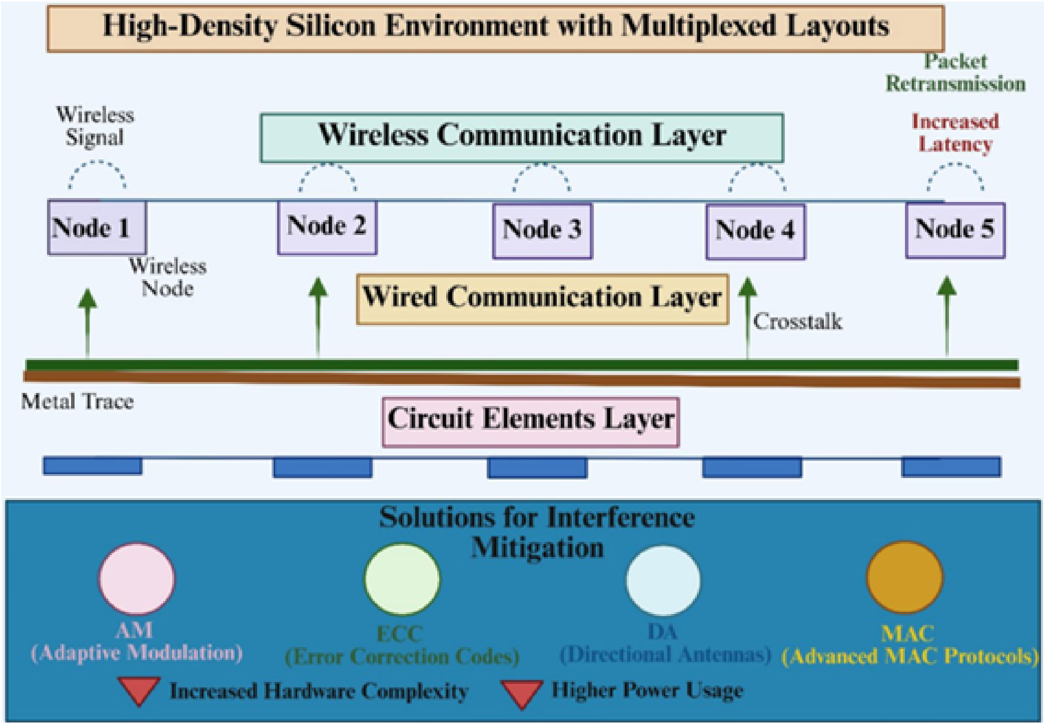

Figure 3: Interference Challenges and MAC-Based Mitigation Techniques

Hybrid MAC approaches combine contention-based and scheduled mechanisms to balance efficiency and adaptability. Such designs allocate dedicated resources for critical traffic while permitting opportunistic access for best-effort services.

Learning-driven MAC protocols demonstrate promising performance improvements in dense and heterogeneous networks. Although these methods enhance adaptability and resource optimization, they introduce computational complexity and require careful training data management. Comparative analysis reveals that performance-driven hybrid and adaptive approaches are better suited for next-generation wireless systems than purely static protocols.

Table 1: Comparison of MAC Protocol Designs for Next-Generation Wireless Systems

Protocol Type	**Latency**	**Throughput**	**Energy Efficiency**	**Scalability**
TDMA	Low	High	Moderate	Limited
CSMA/CA	Moderate	Moderate	Low	High
OFDMA-Based MAC	Very Low	Very High	High	High
AI-Based Adaptive MAC	Very Low	Very High	Very High	Very High
Hybrid MAC	Low	High	High	Moderate

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7 Future Directions

Future MAC protocols will integrate artificial intelligence, edge computing, and dynamic spectrum access to enhance adaptability and efficiency.Future wireless systems are expected to incorporate artificial intelligence natively within network architecture. AI-driven MAC protocols will likely employ predictive analytics to anticipate traffic surges and proactively allocate resources.

The integration of terahertz communication and reconfigurable intelligent surfaces will require new MAC coordination models capable of handling directional beamforming and dynamic spectrum usage.

Edge computing integration may enable decentralized MAC intelligence, allowing local decision-making and reduced latency. Quantum communication research may also influence future MAC security paradigms.

These developments indicate that performance-driven MAC protocol design will remain a critical research domain in next-generation wireless systems.

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8 Conclusion

Performance-driven MAC protocol design is essential for next-generation wireless systems, enabling efficient and reliable communication in complex and dynamic environments.Performance-driven MAC protocol design plays a central role in enabling reliable, scalable, and efficient next-generation wireless communication systems. The increasing diversity of applications and traffic patterns necessitates adaptive, intelligent, and hybrid access strategies. Through careful optimization of throughput, latency, reliability, and energy efficiency, modern MAC protocols can meet the stringent demands of emerging wireless ecosystems.

Although significant progress has been achieved, challenges related to scalability, security, and standardization persist. Continued research and innovation will be essential to fully realize the potential of performance-driven MAC protocols in future wireless networks.

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